

Air University



Object Oriented Programming (Lab)

Year 2025

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Department of Cyber Security

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BS – Cyber Security – Fall – 24 (A)

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Assignment No. 01

Question 01:

Program

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  struct student
6  {
7      string name;
8      int roll_no;
9      float marks;
10 };
11
12 int main ()
13 {
14     student s1;
15
16     cout << "Enter Student's Name: " << endl;
17     getline(cin, s1.name);
18
19     cout << "Enter Student's Roll No: " << endl;
20     cin >> s1.roll_no;
21
22     while (s1.roll_no <= 0)
23     {
24         cout << "Enter Valid Roll No: " << endl;
25         cin >> s1.roll_no;
26     }
27
28     cout << "Enter Student's Marks (in %): " << endl;
29     cin >> s1.marks;
30     while (s1.marks < 0.0)
31     {
32         cout << "Enter Valid Marks Percentage: " << endl;
33         cin >> s1.marks;
34     }
35
36     cout << "\nStudent's Details: " << endl;
37     cout << "Name: " << s1.name << endl;
38     cout << "Roll No: " << s1.roll_no << endl;
39     cout << "Marks: " << s1.marks << "%" << endl;
40
41     return 0;
42 }
```

Output

```
Enter Student's Name:
Usama Arshad
Enter Student's Roll No:
247141
Enter Student's Marks (in %):
77.7

Student's Details:
Name: Usama Arshad
Roll No: 247141
Marks: 77.7%
```

Question 02:

Program

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  struct student
6  {
7      string name;
8      int roll_no;
9      float marks;
10 };
11
12 void display(const student &s)
13 {
14     cout << "\nStudent's Details: " << endl;
15     cout << "Name: " << s.name << endl;
16     cout << "Roll No: " << s.roll_no << endl;
17     cout << "Marks: " << s.marks << "%" << endl;
18     cout << endl;
19 }
20
21 int main ()
22 {
23     student s1;
24 }
```

```
25     cout << "Enter Student's Name: " << endl;
26     getline(cin, s1.name);
27
28     cout << "Enter Student's Roll No: " << endl;
29     cin >> s1.roll_no;
30     while (s1.roll_no <= 0)
31     {
32         cout << "Enter Valid Roll No: " << endl;
33         cin >> s1.roll_no;
34     }
35
36     cout << "Enter Student's Marks (in %): " << endl;
37     cin >> s1.marks;
38     while (s1.marks < 0.0)
39     {
40         cout << "Enter Valid Marks Percentage: " << endl;
41         cin >> s1.marks;
42     }
43
44     display (s1);
45
46     return 0;
47 }
```

Output

```
Enter Student's Name:
Usama Arshad
Enter Student's Roll No:
247111
Enter Student's Marks (in %):
77.5

Student's Details:
Name: Usama Arshad
Roll No: 247111
Marks: 77.5%
```

Question 03:**Program**

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  struct student
6  {
7      string name;
8      int roll_no;
9      float marks;
10 };
11
12 void display(const student s[], int count)
13 {
14     for (int i = 0; i < count; i++)
15     {
16         cout << "\nStudent " << (i+1) << " Details: " << endl;
17         cout << "Name: " << s[i].name << endl;
18         cout << "Roll No: " << s[i].roll_no << endl;
19         cout << "Marks: " << s[i].marks << "%" << endl;
20         cout << endl;
21     }
22 }
```

```
23
24 int main ()
25 {
26     const int max_students = 3;
27     student s1 [max_students];
28
29     for (int i = 0; i < max_students; i++)
30     {
31         cout << "Enter Student " << (i+1) << " Name: " << endl;
32         getline(cin, s1[i].name);
33
34         cout << "Enter Student " << (i+1) << " Roll No: " << endl;
35         cin >> s1[i].roll_no;
36         while (s1[i].roll_no <= 0)
37         {
38             cout << "Enter Valid Roll No: " << endl;
39             cin >> s1[i].roll_no;
40         }
```

```
41
42     cout << "Enter Student " << (i+1) << " Marks (in %): " << endl;
43     cin >> s1[i].marks;
44     while (s1[i].marks < 0.0)
45     {
46         cout << "Enter Valid Marks Percentage: " << endl;
47         cin >> s1[i].marks;
48     }
49 }
50
51 display (s1, max_students);
52
53 return 0;
54 }
```

Output

```
Enter Student 1 Name:
Usama Arshad
Enter Student 1 Roll No:
247141
Enter Student 1 Marks (in %):
77.7
Enter Student 2 Name:
Hamza Ali
Enter Student 2 Roll No:
247127
Enter Student 2 Marks (in %):
77.5
Enter Student 3 Name:
Asad Farooq
Enter Student 3 Roll No:
247171
Enter Student 3 Marks (in %):
77.0
```

```
Student 1 Details:
Name: Usama Arshad
Roll No: 247141
Marks: 77.7%
```

```
Student 2 Details:
Name: Hamza Ali
Roll No: 247127
Marks: 77.5%
```

```
Student 3 Details:
Name: Asad Farooq
Roll No: 247171
Marks: 77%
```

Question 04:**Program**

```
1  #include <iostream>
2  #include <string>
3
4  using namespace std;
5
6  class Book
7  {
8  public:
9      string title;
10     string author;
11     float price;
12 };
13
14 void displayBookDetails(const Book &b)
15 {
16     cout << "\nBook Details:\n";
17     cout << "Title: " << b.title << endl;
18     cout << "Author: " << b.author << endl;
19     cout << "Price: $" << b.price << endl;
20     cout << endl;
21 }
22
23 int main() {
24
25     Book myBook;    // Book Object
26
27     // Take input from the user
28     cout << "Enter book title: ";
29     getline(cin, myBook.title);
30
31     cout << "Enter author name: ";
32     getline(cin, myBook.author);
33
34     cout << "Enter price: ";
35     cin >> myBook.price;
36     while (myBook.price < 0)
37     {
38         cout << "Enter Valid Price: ";
39         cin >> myBook.price;
40     }
41
42     displayBookDetails(myBook);
43
44     return 0;
45 }
```

Output

```
Enter book title: Applied physics  
Enter author name: Isaac Newton  
Enter price: 2300
```

```
Book Details:  
Title: Applied physics  
Author: Isaac Newton  
Price: $2300
```

THANK YOU FOR YOUR TIME!

THE END